IMMUTABLE CANONICAL MANIFEST - LINKED PROVENANCE CHAIN

Canonical root: MACACHOR_COMPRESSION_INVARIANT_v1.0.0_IMMUTABLE.md
Root Version: 1.0.0
Root Date: 2026-08-26
Root SHA-256:
`ec4b2b134e1666a5fb3906cf6a3132349d8c281027c6a61e76ad34cd854b2e5e`

This extension: MACACHOR_SCALAR_REFINEMENT_v3.0_HARDENED
Extension Version: 1.0.1 (Linked Derivation)
Date: 2026-08-21 (refinement) / 2026-09-03 (linking)
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ORCID: 0009-0008-0100-2856
Refinement SHA-256:
`e54d0fa518a54343eccc03dc142f97fb22d793a02492ca8db130114b78a7b9a0`
File: MACACHOR_SCALAR_REFINEMENT_v3.0_HARDENED_6.docx

Provenance Chain

- v1.0.0 (2026-08-26): Immutable compression invariant - `ec4b2b134e1666a5fb3906cf6a3132349d8c281027c6a61e76ad34cd854b2e5e`
    - Slogan: COMPRESS COMPLEXITY. PRESERVE INVARIANTS. RETAIN PROVENANCE. EXPOSE LOSS.
    - Policy: Exact artifact immutable, amendments create new version + hash
- v1.0.1 (2026-09-03): Hardened mathematical derivation - `e54d0fa518a54343eccc03dc142f97fb22d793a02492ca8db130114b78a7b9a0`
    - Status: Mathematically hardened formulation
    - Resolves v2.0 audit gaps: coherence, invertibility, dynamics, ground-state, background metric, global convergence
    - Explicit separation: Theorem / Proposition / Definition / Axiom / Conjecture / Empirical Interpretation

Lock Policy (Inherited)

- This exact linked artifact is immutable.
- No substantive change may reuse version 1.0.1.
- Every substantive amendment creates new version and new SHA-256 hash.
- Superseded versions remain preserved.
- "Irrevocable" means no retroactive canonical rewriting; does not claim physical impossibility of file alteration.
- Any AI-governance deployment must preserve provenance, uncertainty, auditability, accountability, and material evidence.

Canonical Slogan (Inherited)

COMPRESS COMPLEXITY. PRESERVE INVARIANTS. RETAIN PROVENANCE. EXPOSE LOSS.

Hardened Claims Status - v3.0

Statement	Status	Basis
$\nu = 1/\psi^2$ single-valued on $\psi>0$	Theorem	Axiom 1 positivity
F non-increasing	Theorem / energy identity	Axiom 5 + regularity
$\psi\equiv M$ unique positive zero-energy minimizer	Proposition	Def 6.1 + positivity
$\psi\equiv M$ has C=1	Theorem	Def 3.1, 4.1
$\psi\equiv M$ linearly exponentially stable	Theorem	Linearization + compact spectrum
$\psi=0$ linearly unstable	Theorem	$f'(0)=M^2>0$

Statement	Status	Basis
Every positive initial → M	Open / conditional	Requires global theorem
M physically fundamental	Hypothesis	Not derived
vcrit=6.8 threshold	Empirical interpretation	Requires calibration

Compact Equation Set

$M = (\sqrt{5} - 1)/2$
$\rho = \psi^2$
$v = 1/\psi^2$
$C[\psi] = 1 - \mathrm{Var}(\psi^2)/[]\psi^2[]^2$
$V(\psi) = 1/4(\psi^2 - M^2)^2$
$\partial\psi/\partial t = D\Delta_g\psi - (\psi^2 - M^2)\psi$
$F[\psi] = \int_M [D/2 \, |\nabla\psi|^2 + V(\psi)] \, dV_g$
$dF/dt = -\int_M (\partial\psi/\partial t)^2 \, dV_g \leq 0$
$\psi_0 \equiv M \Rightarrow \rho_0 = M^2, C = 1$
$\partial u/\partial t = D\Delta_g u - 2M^2 u + O(u^2)$

Integrity Statement

This manuscript distinguishes mathematical consequence from modeling choice. M, ψ interpretation, positivity domain, compact boundaryless manifold, and coherence functional are model assumptions unless independently derived. Mathematical results apply to specified model only. No appeal to authority, threshold, notation, or terminology substitutes for proof. Physical reality requires independent experimental validation.

Verification

Root SHA-256 verified:
ec4b2b134e1666a5fb3906cf6a3132349d8c281027c6a61e76ad34cd854b2e5e
Refinement SHA-256:

e54d0fa518a54343eccc03dc142f97fb22d793a02492ca8db130114b78a7b9a0
Linking preserves both hashes without alteration of root artifact.